

Cleaning steps

The purpose of this exercise is to practice data cleaning steps using Excel/LibreOffice Calc to check the data rather than using Jupyter or RStudio.

First, the dataset is clearly not clean, with many records showing the word ERROR and empty cells.

	A	B	C	D	E	F	G	H
1	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
2	TXN_1961373	Coffee	2	2	4	Credit Card	Takeaway	2023-09-08
3	TXN_4977031	Cake	4	3	12	Cash	In-store	2023-05-16
4	TXN_4271903	Cookie	4	1	ERROR	Credit Card	In-store	2023-07-19
5	TXN_7034554	Salad	2	5	10	UNKNOWN	UNKNOWN	2023-04-27
6	TXN_3160411	Coffee	2	2	4	Digital Wallet	In-store	2023-06-11
7	TXN_2602893	Smoothie	5	4	20	Credit Card		2023-03-31
8	TXN_4433211	UNKNOWN	3	3	9	ERROR	Takeaway	2023-10-06
9	TXN_6699534	Sandwich	4	4	16	Cash	UNKNOWN	2023-10-28
10	TXN_4717867		5	3	15		Takeaway	2023-07-28
11	TXN_2064365	Sandwich	5	4	20		In-store	2023-12-31
12	TXN_2548360	Salad	5	5	25	Cash	Takeaway	2023-11-07
13	TXN_3051279	Sandwich	2	4	8	Credit Card	Takeaway	ERROR
14	TXN_7619095	Sandwich	2	4	8	Cash	In-store	2023-05-03
15	TXN_9437049	Cookie	5	1	5		Takeaway	2023-06-01
16	TXN_8915701	ERROR	2	1.5	3		In-store	2023-03-21

Fig. 1, an initial display of the dataset

The first step is to identify missing values and wrong values, and decide whether to replace them or remove those records.

The Total Spent column has some missing values, but it is evident that it follows the formula of Quantity multiplied by Price Per Unit. So, the plan is to use a helper column to store the results and then transfer the results into the actual column.

	A	B	C	D	E	F	G	H	I	J
1	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date		
2	TXN_1961373	Coffee	2	2		Credit Card	Takeaway	2023-09-08		4
3	TXN_4977031	Cake	4	3	12	Cash	In-store	2023-05-16		12
4	TXN_4271903	Cookie	4	1	ERROR	Credit Card	In-store	2023-07-19		4
5	TXN_7034554	Salad	2	5	10	UNKNOWN	UNKNOWN	2023-04-27		10
6	TXN_3160411	Coffee	2	2	4	Digital Wallet	In-store	2023-06-11		4
7	TXN_2602893	Smoothie	5	4	20	Credit Card		2023-03-31		20
8	TXN_4433211	UNKNOWN	3	3	9	ERROR	Takeaway	2023-10-06		9
9	TXN_6699534	Sandwich	4	4	16	Cash	UNKNOWN	2023-10-28		16
10	TXN_4717867		5	3	15		Takeaway	2023-07-28		15
11	TXN_2064365	Sandwich	5	4	20		In-store	2023-12-31		20
12	TXN_2548360	Salad	5	5	25	Cash	Takeaway	2023-11-07		25
13	TXN_3051279	Sandwich	2	4	8	Credit Card	Takeaway	ERROR		8

Fig. 2, using a helper column J to store the values

Using the formula, `=IF(OR(E8="",E8="ERROR",E8="UNKNOWN"),C8*D8,E8)`, and then Ctrl+D, it is possible to obtain those values, and then copy paste the whole column J into column C, but paste only the values rather than the formula.

Apply the same process to Quantity and Price Per Unit but using division instead. Quantity should be Total Spent divided by Price Per Unit, and Price Per Unit should be Total Spent divided by Quantity.

In some cases, both values are missing, making it impossible to make the formulas work.

236	TXN_6334895	Smoothie	1	4	4	Credit Card	ERROR	2023-07-27
237	TXN_7077119	ERROR	1	4	4	Digital Wallet	In-store	2023-12-30
238	TXN_8562645	Salad	ERROR	5	UNKNOWN		In-store	2023-05-18
239	TXN_4700144	Coffee	4	2	8		In-store	2023-12-21

Fig. 3, example from non-cleaned data

236	TXN_6334895	Smoothie	1	4	4	Credit Card	ERROR	2023-07-27
237	TXN_7077119	ERROR	1	4	4	Digital Wallet	In-store	2023-12-30
238	TXN_8562645	Salad	#VALUE!	5	#VALUE!		In-store	2023-05-18
239	TXN_4700144	Coffee	4	2	8		In-store	2023-12-21

Fig. 4, result from using formulas

The best course of action would be to remove these records because not having at least two values out of three means there is either a failure to record sales correctly, or it may be a clerical mistake. This is done by using Autofilter feature.

Next, there are empty cells as well as wrong values in the Item column. Since Item is a categorical identifier and there is no reliable method to infer the missing product name, these records should be removed.

After clearing this data, the next step is to look into other possible missing or wrong values in other columns. The columns Transaction Date and Location have missing value, however, it is reasonable to change the values to "Unknown" because these records can still be used for building charts. The bigger concern at this point is the ERROR value in the column Payment Method. The author of the dataset did not specify what these values could mean as there are multiple interpretations, such as whether the payment was denied due to card payment denied, or whether these transactions have not been recorded correctly at the time of input.

The records were removed because the meaning of the ERROR value could not be established, making it impossible to determine whether these transactions represented successful sales or failed transactions. Including them could introduce bias into payment method analysis.

	A	B	C	D	E	F	G	H
1	Transaction ▼	Item ▼	Quantity ▼	Price Per Unit ▼	Total Spent ▼	Payment Method ▼	Location ▼	Transaction Date ▼
34	TXN_6688524	Coffee	4	2	8	ERROR		2023-06-29
46	TXN_6342161	Salad	5	5	25	ERROR	Takeaway	2023-01-08
82	TXN_8268061	Salad	3	5	15	ERROR	Takeaway	2023-08-20
91	TXN_6955416	Salad	4	5	20	ERROR	In-store	2023-02-25
136	TXN_5265095	Coffee	1	2	2	ERROR		2023-06-23

Fig. 5, records with ERROR value in payment method

After removing the ERROR records for payment method, there are still empty cells and "UNKNOWN" values, so they should be replaced with **"Not recorded"**. The reason is because it is reasonable to conclude that these transactions are legitimate. For example, when the value is unknown for payment method, it could entail that the payment was done either through cash or card, and empty cells can indicate that a real transaction was made but could not register the type. This replacement of value was done using Ctrl+H to find and replace. To represent the empty cell, the specific combination of characters is ^\$.

The value **"Not recorded"** was also used to replace the wrong values for Location.

The Transaction Date column contained missing values. These records were retained because the absence of a date does not invalidate the transaction itself and for data integrity purposes to not convert the column into text type. These records should still be excluded from time-based charts.

The assumption of using replacement for these records is that it should be possible to trace back the transactions and be able to correct these records.

After all these steps, the dataset records went from 10,000 to 8,709 . The next step is to identify any duplicates.

Conditional Formatting for A1:A1048576

Conditions

Condition 1

Formula is

Apply Style: **Transaction ID**

Add Delete Up Down

Cell Range

Range:

Help OK Cancel

Fig. 6, conditional formatting to identify duplicates

Conditional formatting was applied to the dataset to identify identical records across all columns. No duplicate rows were detected.